
Survey Design and Questionnaire Construction

Advanced Topics: Methodology, Philosophy, and Research Design

1. Survey Design Overview

Survey = standardized survey of a sample to draw inferences about a population.

Cross-sectional vs. longitudinal (panel, trend). Postal, telephone, online, face-to-face.

Total Survey Error (Groves 2004): Sampling + coverage + nonresponse + measurement error.

2. Questionnaire Construction

Basic rules: Simple, short, unambiguous. No double negatives. No suggestive questions. No double-barreled items.

Scales: Likert (1-5 or 1-7), semantic differential, visual analog scale.

Item polarity: Alternate positive and negative items (avoid acquiescence).

3. Measurement Error and Response Biases

Social desirability: Answering in socially approved direction. Mitigation: anonymity, neutral wording.

Acquiescence: Agreement tendency regardless of content. Mitigation: balanced items.

Extreme response style: Using only endpoints. Mitigation: more categories, forced choice.

Primacy/recency effect: First/last option chosen disproportionately.

4. Pretest and Pilot Study

Cognitive pretest: Test persons asked how they understand questions. Think-aloud protocols.

Pilot study: Complete dry run with n=50-100. Item analysis: difficulty, discrimination, reliability.